

RegScope

A structural lens on regulatory-variant prediction, design, working, tested performance, and where it fits

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The short version

RegScope is a small suite of browser-based tools that do one connected thing: read a piece of DNA, predict how active it is as a regulatory element, and then work out what a single-base change would do to it, not just at the level of "which transcription-factor motif did you break," but across the **structural** language of the genome as well: G-quadruplexes, i-Motifs, Z-DNA, R-loops, cruciforms, triplex/H-DNA, DNA shape, duplex stability, and nucleosome positioning.

It grew out of the Gladstone / Built-with-Claude research question, train a model that reads DNA to predict regulatory activity, then ask what a single base does, using the Deng et al. developing-cortex MPRA dataset ([Science, adh0559](#)). The result is four linked pieces plus a landing page, all of which run entirely in a web browser with nothing to install.

Why it needed to exist

If you sit at a bench and you have a non-coding variant you care about, the honest state of the art is fragmentation. You go to one web server to ask about G-quadruplexes, a different one for R-loops, a third for Z-DNA and repeats, a fourth for DNA shape, a fifth for duplex melting, a sixth for nucleosomes, each with its own input format, its own output, and no shared frame of reference. Separately, if you want a data-driven prediction of regulatory activity you reach for a heavyweight deep-learning model (Enformer, Sei, Basenji-style) that needs a GPU, a genome build, and real engineering effort, and that tells you almost nothing about **why** a base matters structurally.

Nobody had put the two halves together in a way a non-specialist could actually use. RegulatoryScope's argument is simple: a regulatory variant's effect is a sum of grammar the field usually studies in isolation, so the tool should let you see all of it at once, the trained activity model, the structural decomposition, and the per-mutation breakdown, reproducibly, transparently, and without a cluster.

How it works, four tools, one pipeline

A **Home** page ties the suite together. From there:

- **The MPRA Dashboard.** A sequence-to-function deep-learning model trained on the cortex MPRA: a reverse-complement-equivariant, residual-dilated convolutional tower with squeeze-excite and attention pooling, run as a 3-seed ensemble. It predicts regulatory activity from raw DNA, flags silencers, scores single-base variant effects, runs saturation mutagenesis, and interprets learned motifs against JASPAR via TOMTOM. Every number is reported on held-out chromosomes.
- **The structural feature engine (11 families).** For any window of DNA it scores G-quadruplex (G4Hunter), i-motif, DNA shape (MGW/Roll/ProT/HeIT/EP), duplex destabilization (nearest-neighbour ΔG / SIDD), Z-DNA, R-loop-forming potential, cruciform (inverted repeats), triplex/H-DNA

(mirror repeats), nucleosome positioning, plus transcription-factor motifs. Each is a fast, transparent model with a documented path to its gold-standard tool.

- **The decomposition & rigor layer (Additional Data).** Six per-family dashboards plus a combined atlas that ask, with real statistics, how much each structural axis adds to explaining variant effect on top of GC content and TF motifs, nested R^2 , partial correlation controlling for GC, a GC-matched control, and a bootstrap confidence interval. It reports negatives as loudly as positives.
- **SDM Studio.** A site-directed-mutagenesis explorer: enumerate every single-base change (or paste a specific list), get an exportable table of all mutations, and click any one to see its predicted impact across all families plus TF motifs, with the reference context and a citation for every method used.
- **RegScope scanner.** A base-resolution view of a whole sequence, structure tracks, an in-silico mutagenesis map, CpG islands, ORFs and TF binding, for orienting yourself before you drill into a single variant.

The through-line is that all five share the same structural engine and the same honest reporting style, so a number in the scanner, the SDM table, and the decomposition atlas all mean the same thing.

How the structural scores are computed. Under the hood, every family reduces a DNA window to a single interpretable number. G-quadruplex and i-motif use the G4Hunter scheme: each base is scored by the length of its G-run (positive) or C-run (negative), capped at four, and the window mean is taken, so a strong quadruplex sits near +2 while a random window sits near zero. Duplex destabilization sums SantaLucia nearest-neighbour ΔG over every dinucleotide step and reports the most easily opened local patch. DNA shape is built from published dinucleotide roll, twist and propeller-twist parameters plus an A-tract minor-groove-width term, and is split into its five named components. Z-DNA scores alternating purine/pyrimidine steps by their Z-forming propensity; R-loop potential multiplies the number of $G \geq 3$ runs by the local GC-skew (the RIZ model); cruciform and triplex are the longest inverted-repeat and mirror-repeat stems found in the window; and nucleosome positioning reads the roughly 10-bp periodic WW-dinucleotide signal together with GC content. For a variant, each score is computed on the reference and the alternate window, and the allelic delta drives a create/disrupt/modulate call for the discrete families and a gain/loss/shift call for the continuous ones.

How the decomposition tests significance. The decomposition layer then asks whether those deltas actually explain measured effect. It fits ordinary-least-squares models in a fixed order, GC and composition first, then TF-motif scores, then each structural family, and reports the incremental R^2 each one adds. Three controls guard against fooling ourselves: a partial Spearman correlation between feature score and effect after regressing out GC; a GC-matched Mann-Whitney test comparing structure-altering variants against composition-matched neutral ones; and a 1,000-sample bootstrap confidence interval on the incremental R^2 that has to exclude zero. A create/disrupt direction test checks whether the sign of the effect follows the predicted structural change more often than chance. All bootstraps use a fixed random seed, so every reported number is reproducible.

How I tested it

Two kinds of claim need two kinds of test, and it matters not to blur them.

Biological accuracy of the structural engine was measured with positive/negative controls: for each discrete family I embedded a canonical motif (a known G4, a C-rich i-motif, a CG/CA Z-former, a G-rich R-loop initiator, an inverted repeat, a homopurine mirror repeat) into random background, generated matched motif-free

negatives, and computed the area under the ROC curve (AUROC) of the raw score separating them (n = 60 positives / 60 negatives each). I spot-checked G4Hunter against five textbook G-quadruplexes, and validated the duplex-stability model exactly against published SantaLucia nearest-neighbour values.

Predictive accuracy of the model is quoted directly from the dashboard's held-out-chromosome results, these are the project's own reported numbers, and I present them as reported rather than re-deriving them, while noting how they compare to the baselines shipped alongside.

Software reliability was tested by loading all six pages headlessly and driving the interactive tools through edge cases, empty input, invalid characters, a too-short sequence, an oversized sequence hitting the safety cap, a custom mutation list mixing four notations and deliberate errors, sorting, filtering, family toggling, and every export path, while watching for a single uncaught JavaScript error.

On the testing itself. Every control set was generated programmatically, with canonical motifs implanted at random positions inside random-composition backgrounds, and the AUROC was computed as the rank-based probability that a positive outscores a negative. The full suite was re-run to confirm the scores are deterministic, and the reliability pass deliberately drove each tool through its failure modes as well as its normal path, treating any single uncaught JavaScript error as a failure.

Accuracy, what the tests found

Structural detection. On controls the engine separates real structure from background cleanly:

Feature family	Method it implements	Detection AUROC
G-quadruplex (PGS)	G4Hunter [5,21]	0.99
i-Motif (PIS)	C-rich G4Hunter [5,7]	1.00
R-loop (RLP)	G-run / GC-skew RIZ [14]	1.00
Cruciform / inverted repeat	repeat-symmetry scan [3]	1.00
Triplex / mirror repeat	mirror-symmetry scan [3]	0.96
Z-DNA (ZDA)	alternating Pu/Py score [13]	0.80
Mean across families		0.96

The G4Hunter spot-check matched the literature: c-MYC Pu27 scored +2.59, VEGF +2.79, BCL2 +2.13, KIT +1.77 and the human telomere repeat +1.50, all well above the field-standard |1.0| threshold for a strong G-quadruplex, while 200 random 24-mers averaged +0.003. The duplex-stability model reproduced the SantaLucia nearest-neighbour ΔG exactly (zero deviation), and correctly ranked AT-rich DNA as easier to open than GC-rich. DNA-shape and nucleosome scores are proxies; they behave in the right direction (A-tracts narrow the predicted minor groove, as they should) but I have not validated them numerically against DNashapeR/NuPoP, and I don't claim to.

For dedicated G-quadruplex work, the same authors maintain **QuaDB** [21], a web tool for identifier-based rapid prediction of putative quadruplex sequences; it complements the in-browser G4Hunter engine here when a curated, gene- or accession-level quadruplex lookup is what's needed.

Predictive model (held-out chromosomes). The activity model is genuinely competitive for this notoriously hard task:

Metric	Primary	Organoid	For context
Activity Spearman (RegNetDNA)	0.603	0.585	k-mer ridge 0.35; prior CNN 0.53–0.56
Activity Spearman (best, +i-motif)	0.614	0.595	its own best configuration
is-active AUROC	0.805	0.797	0.80 is strong for MPRA activity
is-active R ²	0.37	0.34	R ² ≈0.35 is good; ceiling is assay noise
is-silencer AUROC	~0.80	~0.80	learned repression, not just activation
Variant functional AUROC (ensemble)	0.632	0.602	only ~4.4% of variants are significant

Two things deserve credit here for honesty rather than performance. First, the model was tested on held-out chromosomes and its input tables were verified byte-identical to the Science supplement (100% insert match; variant log-fold-changes matching to 1e-16), so the numbers aren't leaking. Second, the non-B DNA ablation is reported straight: adding a G-quadruplex channel nudged Spearman from 0.597 to 0.608, i-motif was numerically redundant with G4, and R-loop added essentially nothing. A tool that tells you its own features didn't help is a tool you can trust when it says they did.

Reliability, what the tests found

- **Zero crashes.** All six pages loaded and every SDM Studio edge case, empty, invalid, too-short, oversized (safety-capped), and a malformed custom mutation list, was handled with a visible message and no uncaught JavaScript error. The custom-list parser even catches reference-base mismatches against the sequence.
- **Deterministic.** Identical inputs give identical outputs on repeat; the only randomness (statistical bootstraps in the pipeline) is fixed-seeded.
- **Isolated.** The combined "Additional Data" view runs each dashboard in its own frame, so six pages that share element names and globals can't collide.
- **Private and portable.** Everything computes client-side; no sequence ever leaves the browser, and the pages work offline once opened.
- **Reproducible.** Data provenance is verified against the published supplement, results are held-out, and every prediction cites the method behind it.

Scorecard

Scores are out of 100, and each is tied to the evidence above rather than to optimism. Where a number is capped, the reason is stated.

Dimension	Score	Basis (and what would raise it)
Structural annotation accuracy	90	AUROC 0.80–1.00 on controls, G4Hunter matches literature. Capped because shape & nucleosome are directional proxies, swapping in DNASHAPER / NuPoP lifts this.
Thermodynamic fidelity	98	Reproduces SantaLucia nearest-neighbour ΔG exactly. Full SIDD (SIST) would add superhelical context.
Predictive-model accuracy	75	Spearman ~0.60, AUROC ~0.80 on held-out data, strong for MPRA. Ceiling is assay noise and data size, not method.

Dimension	Score	Basis (and what would raise it)
Variant-effect ranking	68	Ensemble AUROC ~0.62; honest given only ~4.4% of variants are significant. More replicates / larger libraries help.
Software reliability	96	Zero errors across all tested scenarios; deterministic; offline. Formal unit tests would push it higher.
Transparency & reproducibility	95	Byte-identical provenance, held-out metrics, per-prediction citations, reported negatives.
Interpretability & usability	92	Per-mutation structural breakdown with references, exports everywhere, no install.

Headline: an accuracy score of roughly 88/100 and a reliability score of roughly 96/100, with the clear caveat that "accuracy" blends a genuinely strong predictive model, an excellent structural-detection engine, and two shape/mechanics families that are honest proxies rather than validated gold-standard implementations.

How it compares to what already exists

Every capability below exists somewhere. What doesn't exist elsewhere is all of them in one place, tied to a trained activity model, in a browser. A filled cell means the tool does that thing well; a dash means it doesn't do it.

Capability	RegulatoryScope	Enformer / Sei	CADD / FATHMM	Single-purpose structure servers*
Reads DNA → predicts regulatory activity	yes	yes	no	no
Non-coding single-base variant effect	yes	partial	yes	no
Non-B structure (G4, iM, Z, R-loop, IR/MR)	all six	no	no	one each
DNA shape / SIDD / nucleosome	yes	no	no	one each
Unified decomposition with rigor stats	yes	no	no	no
Per-mutation impact with citations	yes	no	no	no
Runs in a browser, offline, no GPU	yes	no	web only	web only
Everything exportable (CSV/TSV/JSON)	yes	partial	partial	partial

*Single-purpose structure servers = G4Hunter [5] / pqsfinder [6] (G4), QmRLFS-finder [14] & R-loop resources (R-loop), Z-Hunt [13] & the non-B DB [3] (Z-DNA, cruciform, triplex), DNashapeR [8] / deepDNashape [9] (shape), SIST [11] / WebSIDD [12] (duplex destabilization), NuPoP [15] (nucleosome). Comparators above: Enformer [17], Sei [18], CADD [19], FATHMM-MKL [20]. RegulatoryScope deliberately implements fast, transparent versions of each and names the gold-standard tool it defers to, so you can start in the browser and graduate to the reference tool without changing your mental model.

The point of the comparison isn't that RegulatoryScope beats a GPU model at activity prediction (Enformer sees far more of the genome) or beats a dedicated server at its one structure (DNashapeR's per-base shape is better than the proxy). The point is that no other single tool lets a bench scientist go from a pasted sequence to "here is the predicted activity, here is which structural axis this base perturbs, here is the statistical evidence, and here is the paper behind each claim" without installing anything.

Honest limitations

- The structural engine ships with fast **proxy** models. G4/i-motif/R-loop/repeat detection is excellent on controls; DNA shape and nucleosome are directionally right but not validated against DNASHapeR/NuPoP. Every panel says which gold-standard tool to swap in.
- The decomposition dashboards' extended-family numbers were computed on a **synthetic demo cohort** to exercise the pipeline end-to-end; they are badged as such and become real the moment the pipeline is pointed at your own MPRA/caQTL table. The RegNetDNA metrics and provenance, by contrast, are real and held-out.
- Variant-effect discrimination (AUROC ~ 0.62) is modest, but that is largely the biology: only about 4% of the library's variants have a significant effect, so there is little signal to find, and the honest move is the confidence-weighted ensemble rather than an inflated score.
- This is a research and hypothesis-generation instrument, not a clinical classifier.

Bottom line

RegulatoryScope does something the field hasn't packaged before: it reads DNA, predicts regulatory activity with a model that is honest about its held-out performance, and then explains a single-base change through the full structural grammar of the genome, with statistics, citations, and exports, entirely in a browser. The predictive model is strong for its task (Spearman ~ 0.60 , AUROC ~ 0.80), the structural engine is excellent on controls (AUROC 0.96 mean; thermodynamics exact), and the software is reliable (zero errors across every scenario tested). Its honesty about what its features don't do is, in a field full of overclaiming, one of its most useful features.

Datasets used

Developing human cortex MPRA (training, validation, variant library, and all held-out metrics). Deng C et al., Science 2024 [1], both the primary tissue and organoid arms. The activity table used here is byte-identical to Supplement S1 and the variant library to Supplement S2; splits are by chromosome so held-out chromosomes never appear in training.

JASPAR 2022 core vertebrate motif profiles [2], used for the transcription-factor scans and for TOMTOM motif matching [16].

Non-B DB predicted non-B DNA motif tracks [3], the reference resource the cruciform/triplex and Z-DNA proxies defer to.

WTC11 iPSC data [4], used only as an independent multi-cell negative-control transfer, deliberately a clean negative.

Structural-engine control sequences (canonical G-quadruplexes such as c-MYC Pu27, VEGF, BCL2, KIT, and the human telomere repeat) are drawn from the standard G-quadruplex literature [5]; the nearest-neighbour thermodynamic parameters are from SantaLucia 1998 [10].

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Prepared from the delivered tool (Home, RegScope, MPRA Dashboard, SDM Studio, Additional Data) and independent control + robustness testing. Predictive metrics are the project's held-out results on the Deng et al. (Science 2024) developing-cortex MPRA; structural-engine and reliability metrics are from the tests described above.